

Bayesian Techniques in Macroeconomics

Paris-Dauphine & PSL Universities
Master - Quantitative Economics, 2nd year

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CHAPTER 2 A Production Economy

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HANDOUT OBJECTIVES

- Solving a rational expectation model;
- Linearizing a system of equations;
- Extending a model and checking its solution.

Session roadmap

Economic intuition. Capital links current investment to future production. **Bayesian/econometric intuition.** The equilibrium solution becomes the transition equation used by the Kalman filter. **Model.**

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Content of the Handout

Derive the steady state, log-linearize, and verify determinacy. **MATLAB/Dynare exercise.** Compare analytical and Dynare solutions, then interpret moments and IRFs. **Takeaway.** Never estimate a DSGE model before residual and Blanchard–Kahn checks pass.

1. A production economy

Our economy is characterized by a representative household which consumes, produces goods, accumulates physical capital involved in the production. There is no labour in this economy as household's labour supply is assumed to be inelastic to macroeconomic fluctuations while technology does not involve hours worked. Households are immortal and infinite, the economy is populated by an infinite number of $j \in [0, 1]$ distributed on an unitary continuum.

The household face the following maximization problem:

$$\begin{aligned} \max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \log(C_{jt+\tau}) \\ s.t. : C_{jt} + K_{jt} = A_t K_{jt-1}^\alpha + (1 - \delta) K_{jt-1} \end{aligned}$$

The budget constraint is actually the result of the combination of the GDP and the accumulation of physical capital. First, the GDP reads as follows:

$$C_{jt} + I_{jt} = Y_{jt}. \quad (1)$$

The total income Y_{jt} is spent on consumption goods C_{jt} and investment goods I_{jt} .

The law of motion of capital reads as in the [Solow \[1956\]](#)'s model:

$$I_{jt} = K_{jt} - (1 - \delta) K_{jt-1} \quad (2)$$

Investment is determined by the gap between current capital K_{jt} and depreciated capital $(1 - \delta) K_{jt-1}$. In the latter expression, the parameter $0 \leq \delta \leq 1$ denotes the depreciation rate of capital.

The standard Cobb-Douglas technology is determined by:

$$Y_{jt} = A_t K_{jt-1}^\alpha \quad (3)$$

where $0 < \alpha \leq 1$ is the share of capital involved in the production process of new goods Y_{jt} while A_t is a productivity shock which temporary affects the society's ability to produce goods, it affects all the households in our economy in the same way. The production combines technology and capital K_{jt-1} bought in $t - 1$ as capital requires one period to be settled (the so called time-to-build in the RBC literature). Concerning the productivity shock, it is assumed to follow a simple AR(1) process given by:

$$\log(A_t) = \rho_A \log(A_{t-1}) + \eta_t^A, \quad (4)$$

where $0 \leq \rho_A < 1$ is the autocorrelation coefficient of the shock and η_t^A is an innovation distributed normally with zero mean and finite variance σ_A^2 such that $\eta_t^A \sim \mathcal{N}(0, \sigma_A^2)$.

Exercise 1: A production economy

Consider the production economy model.

1. Set the Lagrangian problem.
2. Find the first order conditions. Get rid off the Lagrangian multiplier to obtain the Euler equation on physical capital.
3. Write the three equations of the model. Apply aggregation to find the general equilibrium

conditions such that $C_t = \int_0^1 C_{jt} dj$, $K_t = \int_0^1 K_{jt} dj$ and $Y_t = \int_0^1 Y_{jt} dj$. Find the steady state.

4. Linearize the three equations.

2. Solving a linear rational expectation model

2.1 A general solution

Recall that in a general form, a model is a set of nonlinear equations:

$$\mathbb{E}_t [f_\theta(x_{t+1}, x_t, x_{t-1}, \epsilon_t)] = 0, \quad (5)$$

where θ is the set of structural parameters, x_t are endogenous variables, ϵ_t are gaussian innovations and $f_\theta(\bullet)$ is the set of equations.

As discussed in the previous handout, we apply linearization methods to $f_\theta()$ to obtain the set of linear equations:

$$\mathbb{E}_t [\hat{f}_\theta(\hat{x}_{t+1}, \hat{x}_t, \hat{x}_{t-1}, \epsilon_t)] = 0 \quad (6)$$

A classification of variables and equations. In order to avoid singular matrices, we must distinguish between three type of variables: (i) state variables $\hat{s}_t$, (ii) jump variables $\hat{j}_t$ and (iii) exogenous shock processes $\hat{z}_t$.² State variables feature persistence and are determined by other variables, they are key in shaping the propagation mechanism of a business cycle model. Unlike state variables, jump variables don't feature some persistence. Finally, exogenous shocks are simply self-determined variables as they don't depend on any other variables. Note that Dynare automatically detects states and jump variables and reports their respective numbers in the command window of MATLAB.

Given the previous dichotomy across endogenous variables $\hat{x}_t = [\hat{s}_t, \hat{z}_t, \hat{j}_t]'$, we also must distinguish between three types of equations in our linearized model $\hat{f}_\theta()$:

$$0 = E_t \begin{bmatrix} A_0 \hat{s}_t + A_{-1} \hat{s}_{t-1} + B_0 \hat{j}_t + C_0 \hat{z}_t \end{bmatrix}, \quad (7)$$

$$0 = E_t \begin{bmatrix} D_1 \hat{s}_{t+1} + D_0 \hat{s}_t + D_{-1} \hat{s}_{t-1} + M_1 \hat{j}_{t+1} + M_0 \hat{j}_t + L_1 \hat{z}_{t+1} + L_0 \hat{z}_t \end{bmatrix}, \quad (8)$$

$$\hat{z}_t = Z \hat{z}_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q). \quad (9)$$

The first set of equations stacks equations with no expectation terms, the second set of equations must feature at least one forward looking variable, and the last set of equation are exogenous shock processes.

The functional form of the solution. We are looking for a recursive solution of our model:

$$\hat{x}_t = F \hat{x}_{t-1} + G \hat{z}_t, \quad (10)$$

where F and G are yet two unknown matrices. Note that this solution must hold for any transitions of $\hat{x}$ either from $t+1 \rightarrow t$ or $t \rightarrow t-1$ consistently with our linearized system $\hat{f}_\theta()$. Using our classification, this solution can be expressed as two separate processes:

$$\begin{bmatrix} \hat{s}_t \\ \hat{j}_t \end{bmatrix} = \begin{bmatrix} F_s & 0_{n_j} \\ F_j & 0_{n_j} \end{bmatrix} \begin{bmatrix} \hat{s}_{t-1} \\ \hat{j}_{t-1} \end{bmatrix} + \begin{bmatrix} G_s \\ G_j \end{bmatrix} \hat{z}_t \quad (11)$$

²Note that Dynare has a different classification for endogenous variables as exogenous variables are also considered as state ones.

The unknown in the previous system are matrices F_s , F_j , G_s and G_j . Another way to present the solution reads as follows:

$$\hat{s}_t = F_s \hat{s}_{t-1} + G_s \hat{z}_t, \quad (12)$$

$$\hat{j}_t = F_j \hat{s}_{t-1} + G_j \hat{z}_t. \quad (13)$$

Solution of the system. To obtain the unknowns of our transition equation, consider our first set of equations in Equation 7. We must use equations (12-13) to replace $\hat{s}_t$ and $\hat{j}_t$ as follows:

$$\begin{aligned} 0 &= A_0 \hat{s}_t + A_{-1} \hat{s}_{t-1} + B_0 \hat{j}_t + C_0 z_t \\ 0 &= A_0 (F_s \hat{s}_{t-1} + G_s z_t) + A_{-1} \hat{s}_{t-1} + B_0 (F_j \hat{s}_{t-1} + G_j z_t) + C_0 z_t \\ 0 &= \underbrace{[A_0 F_s + A_{-1} + B_0 F_j]}_{\text{condition (1)}} \hat{s}_{t-1} + \underbrace{[A_0 G_s + B_0 G_j + C_0]}_{\text{condition (2)}} z_t \end{aligned}$$

We do exactly the same for the second set of equation in stacked in Equation 8 to obtain:

$$\begin{aligned} 0 &= \underbrace{[D_1 (F_s)^2 + D_0 F_s + M_1 F_j G_s + D_{-1} + M_0 F_j]}_{\text{condition (3)}} \hat{s}_{t-1} \\ &\quad + \underbrace{[D_1 F_s G_s + D_0 G_s + M_1 F_j G_s + M_0 G_j + D_1 G_s Z + M_1 G_j Z + L_1 Z + L_0]}_{\text{condition (4)}} \hat{z}_t, \end{aligned}$$

Note that we have 4 conditions and four unknown, so we should theoretically be able to derive the value F_s , F_j and G_s , G_j . However in practice, it's not as obvious because condition (3) is a quadratic matrix equation problem of the form $AX^2 + BX + C = 0$. There are two implications from this quadratic problem:

1. Obtaining the closed-form expression of $(F_s)^2$ is rather difficult, but fortunately Uhlig et al. [1995] provides a solution in his Theorem 3. In addition, Dynare has also its own routine to solve linear model (based on the Schur decomposition).
2. The quadratic form of the equations allows multiple roots, that is to say there are two matrices F_s that solves our models. So our economy can possibly follow two different trajectories for the same system of equation. In practice, one of these two path is unstable, so macroeconomists typically check the eigen values of both systems and pick the stable root. Note that Dynare also automatically picks the stable roots and provide to the user the eigen values of the system. Finally, a last remark, you may have noticed that if D_1 is all of zeros, then the quadratic problems disappears.

Let us assume that we have followed carefully theorem (3) from Uhlig et al. [1995], which allowed us to solve the quadratic matrix equation problem and pick the stable root of the system. If F_s is known, then condition (1) allows us to find F_j as follows:

$$F_j = -(B_0)^{-1} (A_0 F_s + A_{-1}) \quad (14)$$

Mixing both condition (2) and (4) allows to pin down the two remaining unknown matrices:

$$G_j = (M_0 + M_1 Z)^{-1} \left[(A_0)^{-1} (B_0 G_j + C_0) (D_1 F_s + D_0 + M_1 F_j + D_1 Z) - (L_1 Z + L_0) \right], \quad (15)$$

$$G_s = -(A_0)^{-1} (B_0 G_j + C_0). \quad (16)$$

Prior the spread of Dynare within the profession, macroeconomists used to linearize their model's equations, to find the steady state, and to fill matrices in [Equation 7-8-9](#) to get the model's dynamics. Changing a single equation would imply to update all the equations. Dynare really simplified the problem by automating the linearization of equations, filling the matrices, find the solution, pick the stable root, plot the IRF, and get second moment statistics. Note also that the number of elements in matrices increases exponentially in the number of variables, this is referred to as [the curse of dimensionality](#). So, most of DSGE models were rather small prior the use of Dynare.

Dynare formulation. Note that Dynare consider exogenous shock processes as state variables, replacing $\hat{z}_t$ into the system, we obtain:

$$\begin{bmatrix} \hat{s}_t \\ \hat{j}_t \end{bmatrix} = \begin{bmatrix} F_s & 0_{n_z} & 0_{n_j} \\ F_j & 0_{n_z} & 0_{n_j} \end{bmatrix} \begin{bmatrix} \hat{s}_{t-1} \\ \hat{j}_{t-1} \end{bmatrix} + \begin{bmatrix} G_s \\ G_j \end{bmatrix} Z \hat{z}_{t-1} + \begin{bmatrix} G_s \\ G_j \end{bmatrix} \epsilon_t, \quad (17)$$

which is formulated in Dynare as:

$$\begin{bmatrix} \hat{s}_t \\ \hat{z}_t \\ \hat{j}_t \end{bmatrix} = \begin{bmatrix} F_s & G_s Z & 0_{n_j} \\ 0_{n_s} & Z & 0_{n_j} \\ F_j & G_j Z & 0_{n_j} \end{bmatrix} \begin{bmatrix} \hat{s}_{t-1} \\ \hat{z}_{t-1} \\ \hat{j}_{t-1} \end{bmatrix} + \begin{bmatrix} G_s \\ 1_{n_z} \\ G_j \end{bmatrix} \epsilon_t \quad (18)$$

Or equivalently to the initial form of the policy rule:

$$\hat{x}_t = F \hat{x}_{t-1} + G \epsilon_t, \quad (19)$$

2.2 An application to the production economy

Let's go back to a production economy linearized as follows:

$$\begin{aligned} -\hat{k}_t + k_1 \hat{a}_t + k_2 \hat{k}_{t-1} - k_3 \hat{c}_t &= 0 \\ -\hat{c}_t + \hat{c}_{t+1} - c_1 \hat{a}_{t+1} - c_2 \hat{k}_t &= 0 \\ -\hat{a}_t + \rho_A \hat{a}_{t-1} + \eta_t^A &= 0 \end{aligned}$$

where $k_1 = K^{\alpha-1} \rho_A$, $k_2 = K^{\alpha-1} \alpha + (1 - \delta)$, $k_3 = \bar{C}/\bar{K}$, $c_1 = \beta \alpha \bar{K}^{\alpha-1}$, $c_2 = \beta \alpha \bar{K}^{\alpha-1} (\alpha - 1)$ and $E_t \hat{a}_{t+1} = \rho_A \hat{a}_t$. Note that this system is referred to as [Equation 6](#) in the previous subsection.

Our solution reads as:

$$\hat{c}_t = F_c \hat{k}_{t-1} + G_c \hat{a}_t, \quad (20)$$

$$\hat{k}_t = F_k \hat{k}_{t-1} + G_k \hat{a}_t. \quad (21)$$

where F_c , F_k , G_c , G_k are yet unknown that we must find. To obtain these unknowns, we must inject these two equations in the previous system. Let us start with the first equation:

$$\begin{aligned} -\hat{k}_t + k_1 \hat{a}_t + k_2 \hat{k}_{t-1} - k_3 \hat{c}_t &= 0, \\ -\left(F_k \hat{k}_{t-1} + G_k \hat{a}_t\right) + k_1 \hat{a}_t + k_2 \hat{k}_{t-1} - k_3 \left(F_c \hat{k}_{t-1} + G_c \hat{a}_t\right) &= 0, \\ \underbrace{[k_2 - k_3 F_c - F_k] \hat{k}_{t-1}}_{\text{condition (1)}} + \underbrace{[k_1 - k_3 G_c - G_k] \hat{a}_t}_{\text{condition (2)}} &= 0. \end{aligned}$$

$$\begin{aligned} -F_k \hat{k}_{t-1} - G_k \hat{a}_t + k_1 \hat{a}_t + k_2 \hat{k}_{t-1} - k_3 F_c \hat{k}_{t-1} - k_3 G_c \hat{a}_t &= 0, \\ (-F_k + k_2 - k_3 F_c) \hat{k}_{t-1} + (k_1 - k_3 G_c - G_k) \hat{a}_t &= 0, \end{aligned}$$

Note that we must keep injecting these equations until we obtain a system made of $\hat{k}_{t-1}$ and $\hat{a}_t$. Following those steps, we obtain two conditions on parameters. For the second equation, we do exactly the same as the previous equation to obtain a system with variables $(\hat{k}_{t-1}, \hat{a}_t)$:

$$\begin{aligned} -\hat{c}_t + \hat{c}_{t+1} - c_1 \hat{a}_{t+1} - c_2 \hat{k}_t &= 0, \\ -\hat{c}_t + F_c \hat{k}_t + G_c \hat{a}_{t+1} - c_1 \hat{a}_{t+1} - c_2 \hat{k}_t &= 0, \\ -\hat{c}_t + (F_c - c_2) \hat{k}_t + \rho (G_c - c_1) \hat{a}_t &= 0, \\ F_c \hat{k}_{t-1} - G_c \hat{a}_t + (F_c - c_2) (F_k \hat{k}_{t-1} + G_k \hat{a}_t) + \rho (G_c - c_1) \hat{a}_t &= 0, \\ \underbrace{[-F_c + (F_c - c_2) F_k] \hat{k}_{t-1}}_{\text{condition (3)}} + \underbrace{[(F_c - c_2) G_k + \rho (G_c - c_1) - G_c] \hat{a}_t}_{\text{condition (4)}} &= 0. \end{aligned}$$

If we find the values of F_c , G_c , F_k , G_k that satisfies conditions (1-4), then we have found the solution of our model. Consider these four conditions:

$$\begin{aligned} \text{condition (1)} : 0 &= k_2 - k_3 F_c - F_k \\ \text{condition (2)} : 0 &= k_1 - k_3 G_c - G_k \\ \text{condition (3)} : 0 &= F_c + (F_c - c_2) F_k \\ \text{condition (4)} : 0 &= -G_c + (F_c - c_2) G_k + \rho (G_c - c_1) \end{aligned}$$

Combining condition (1) and (3), we obtain the following problem:

$$-k_3 F_c^2 + (k_3 c_2 - 1 + k_2) F_c - k_2 c_2 = 0.$$

Note that this problem has two roots, the usual practice in macroeconomics is to pick the 'stable' root of the sytem, ie $|F_c| < 1$. Based on the stable root, we can get the rest of the unknown recursively as follows:

$$\begin{aligned} F_k &= k_2 - k_3 F_c \\ G_c &= \frac{(F_c - c_2) k_1 - \rho c_1}{(F_c - c_2) k_3 + (1 - \rho)} \\ G_k &= k_1 - k_3 G_c \end{aligned}$$

To get a presentation of the solution close to the one of Dynare, the problem can be recasted as follows

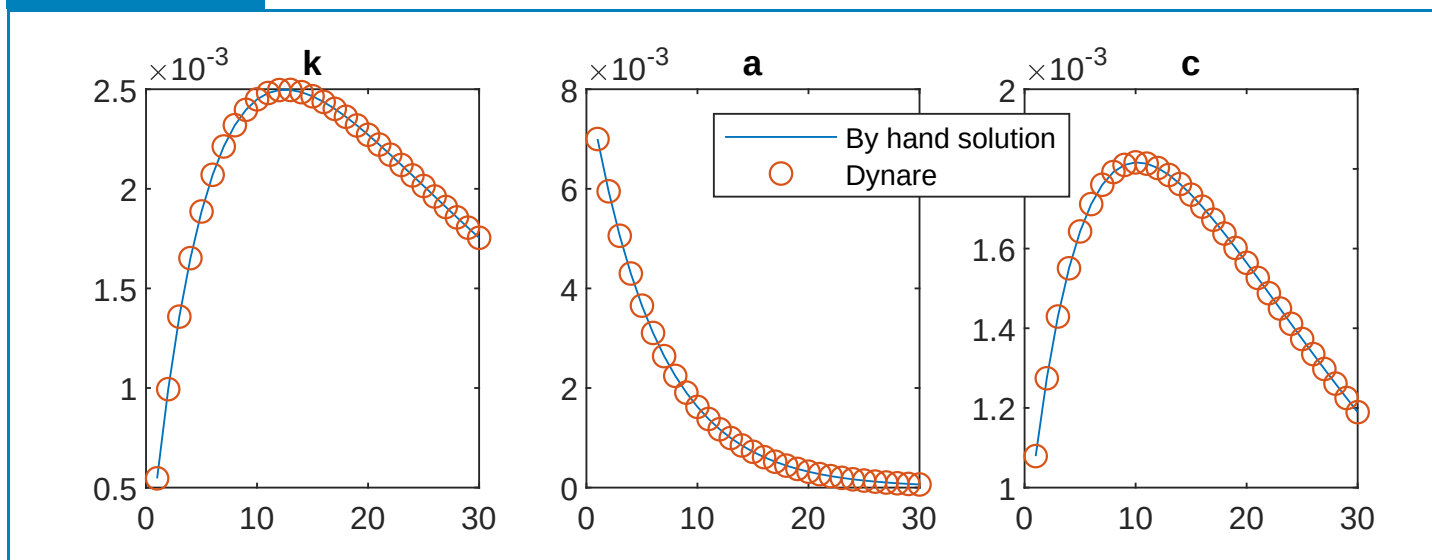
$$\begin{bmatrix} \hat{k}_t \\ \hat{a}_t \\ \hat{c}_t \end{bmatrix} = \begin{bmatrix} F_k & G_k \rho_A & 0 \\ 0 & \rho_A & 0 \\ F_c & G_c \rho_A & 0 \end{bmatrix} \begin{bmatrix} \hat{k}_{t-1} \\ \hat{a}_{t-1} \\ \hat{c}_{t-1} \end{bmatrix} + \begin{bmatrix} G_k \\ 1 \\ G_c \end{bmatrix} \epsilon_t. \quad (22)$$

Which we will write as in the next section:

$$x_t = F x_{t-1} + G \epsilon_t, \quad (23)$$

where $x_t = [\hat{k}_t, \hat{a}_t, \hat{c}_t]'$.

MATLAB box 1 `compareRBC.m`



3. Dynare Programming

We want to write the model into a form Dynare can interpret. In the first part of the Dynare code, we need to define the endogenous variables which appear in the model. There are three equations, so we have three variables: c_t , k_t and a_t . The syntax for writing this in Dynare is `var [variable1 variable2, ... ];` with the line ended by a semicolon. For our model, we write:

```
1 var k a c;
```

Next we write the parameters of the model. In our model, there are four structural parameters (α , β , δ and ρ_A) plus two steady state variables present in the linear model ($\bar{C}$ and $\bar{K}$), giving six parameters to declare. The syntax for writing this in Dynare is `parameters parameter1 parameter2, ... ;` with the line ended by a semicolon. Here we note:

```
1 parameters alpha beta delta rho_A Css Kss;
```

Next, we write the exogenous variables in the model. We only have one: η_t . The syntax for writing this in Dynare is `varexo variable1 variable2, ... ;` with the line ended by a semicolon. For our set-up, we write:

```
1 varexo e_a;
```

Finally, we give the parameters values using our calibration in the parameter block below. The syntax for assigning parameter values is `parameter = value;` which is similar to normal Matlab code. We write:

```
1 alpha      = 0.40;
2 beta       = 0.99;
3 delta      = 0.025;
4 rho_A      = 0.95;
5 Kss        = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
6 Css        = Kss^alpha - delta*Kss;
```

We now write the equations of the linearised model. Those equations are usually written in the same way as they appear in the paper. The equation section starts with `model (linear);` and ends with `end;`.³ A few syntax considerations:

1. Variables which are multiplied (divided) by a parameter are separated by the parameter by the symbol `*` (`/`).

³Note that the tag `model (linear);` includes the option 'linear' because the model is already linearized. For fully non-linear model, just write `model ;`.

-
2. Contemporaneous variables do not have a time subscript in the Dynare code. Lagged variables x_{t-i} are written as `x(i)`, while expectational variables $\mathbb{E}_t x_{t+i}$ are written as `x(+i)`.
 3. Equations are ended with a semicolon ;.

The equations in Dynare syntax then are:

```
1 model(linear);
2     [name='Budget constraint']
3     k = Kss^(alpha-1)*a + (Kss^(alpha-1)*alpha + (1-delta))*k(-1) - Css/Kss*c;
4     [name='Euler equation']
5     c = c(+1) - beta*alpha*Kss^(alpha-1)*(a(+1) + (alpha-1)*k);
6     [name='Productivity shock']
7     a = rho*a(-1)+e_a;
8 end;
```

After specifying the equations, we define the shock, and set its standard deviation. We begin the section with a tag `shocks;` and end it with `end;`. The syntax for specifying each shock in Dynare is `var shock = shock variance;`. For the model, the section is:

```
1 shocks;
2     var e_a; stderr .007;
3 end
```

Next, we can perform a diagnostic of the model stationarity with respect to the Blanchard-Kanh condition. In Dynare, we add:

```
1 check;
```

Finally, we solve the model and perform stochastic simulations. To do so, we employ the function `stoch_simul(options=value) variable 1 variable 2;`. Here we want a 30-period system response of consumption, capital and productivity after a technological shock. We write accordingly:

```
1 stoch_simul(irf=30) c k a;
```

There many options available for the `stoch_simul()` function, see directly the Dynare user guide [Adjemian et al. \[2011\]](#) for details about options.

3.1 Simulating the economy

To run the model, the code has to be saved in a .mod file (mine is called basicRBC.mod). To run the file, you also need to set Matlab's current directory where your Dynare file is. To initiate Dynare, in Matlab's command line, type:

```
console
```

```
dynare basicRBC
```

It is possible to insert more argument in the Dynare command (see the manual for details). After pressing enter, you will then see Dynare at work. What it is doing is reading in the .mod contents, recognising the model's details, and solving for the linear rational expectations solution. A key advantage of Dynare compared to other alternatives is that Dynare creates and fills the matrices with the appropriate parameters to solve the linear rational expectation model and prints and saves all results. A lot of informations pops up in Matlab command window, which concerns the stability of the system and the second order moment statistics of the model.

```
console response
```

```
Using 64-bit preprocessor  
Starting Dynare (version 4.6.2).  
Calling Dynare with arguments: none  
Starting preprocessing of the model file ...  
Found 3 equation(s).  
Evaluating expressions...done  
Computing static model derivatives (order 1).  
Computing dynamic model derivatives (order 2).  
Processing outputs ...  
done  
Preprocessing completed.
```

The first bit of printed output Dynare generates is processing information.

Next, Dynare checks whether the Blanchard-Kanh condition is verified.⁴

⁴This is a condition defining the stability condition of the model. As explained in Dynare website, the Blanchard and Kahn conditions state that in order to have a unique stable trajectory you need to have as many eigenvalues larger than 1 in modulus as you have forward looking variables (variables with leads in Dynare; if leads is on two periods as in $y(+2)$, it counts for 2 forward looking variables).

In addition, a matrix that pops up in the computation of the solution must have full rank.

These conditions can be broken two different ways: with eigenvalues more or less numerous than the forward looking variables. This leads respectively to nonexistence of a stable trajectory or to an infinity of stable trajectories (indeterminacy). Dynare will only proceed with computation if there exists a unique stable trajectory. You can display the eigenvalues with the "check" command added to the line after "steady" (the command to compute the steady state of a model). If this condition does not hold, Dynare stops processing the code and prints an error message.

console response

EIGENVALUES:

Modulus Real Imaginary

0.85 0.85 0

0.9691 0.9691 0

1.042 1.042 0

1.262e+17 1.262e+17 0

There are 2 eigenvalue(s) larger than 1 in modulus
for 2 forward-looking variable(s)

The rank condition is verified.

Here, as the condition is verified, the simulation can continue.

Next, Dynare generates the variance-covariance matrix of the shocks. The elements on the diagonal correspond to the specified values of the shock's variance. Here, we only have one shock, so the matrix is 1x1:

console response

MATRIX OF COVARIANCE OF EXOGENOUS SHOCKS

Variables e_a

e_a 0.000049

Next, we obtain the policy and transition functions. Recall that in a compact form, Dynare computes the solution of the log-linear model (called policy function) as follows:

$$\hat{x}_t = F\hat{x}_{t-1} + G\epsilon_t, \quad (24)$$

where $\hat{x}_t = [\hat{k}_t, \hat{a}_t, \hat{c}_t]$ is the vector of the endogenous variables which is a self-determined process based on previous realizations of $\hat{x}_{t-1}$ + the current realization of shocks $\epsilon_t = [\eta_t^A]$. The matrix A corresponds to the first two rows while B corresponds to the last row of the policy transition matrix.

console response

POLICY AND TRANSITION FUNCTIONS

	c	k	a
k(-1)	0.653600	0.969086	0
a(-1)	0.269310	0.066465	0.950000
e_a	0.283484	0.069963	1.000000

We can then rewrite our backward looking process as:

$$\begin{bmatrix} \hat{c}_t \\ \hat{k}_t \\ \hat{a}_t \end{bmatrix} = \begin{bmatrix} 0 & 0.653600 & 0.269310 \\ 0 & 0.969086 & 0.066465 \\ 0 & 0 & 0.950000 \end{bmatrix} \begin{bmatrix} \hat{c}_{t-1} \\ \hat{k}_{t-1} \\ \hat{a}_{t-1} \end{bmatrix} + \begin{bmatrix} 0.283484 \\ 0.069963 \\ 1 \end{bmatrix} [\eta_t]$$

Since consumption $\hat{c}_t$ is only forward looking in the Euler equation, it does not appear in the policy matrix as its row is only a vector of all zeros.

The next section of Dynare output reports the moments of the model, using the model solution and the specified standard errors of the exogenous shocks.

console response

THEORETICAL MOMENTS

VARIABLE	MEAN	STD. DEV	VARIANCE
c	0.0000	0.0098	0.0001
k	0.0000	0.0135	0.0002
a	0.0000	0.0133	0.0002

Dynare then reports a matrix of theoretical correlations across the endogenous variables. These values are constructed from the variance-covariance matrix of the endogenous variables, which Dynare does not output to the command window.

console response

MATRIX OF CORRELATIONS

Variables	c	k	a
c	1.0000	0.9923	0.5435
k	0.9923	1.0000	0.4351
a	0.5435	0.4351	1.0000

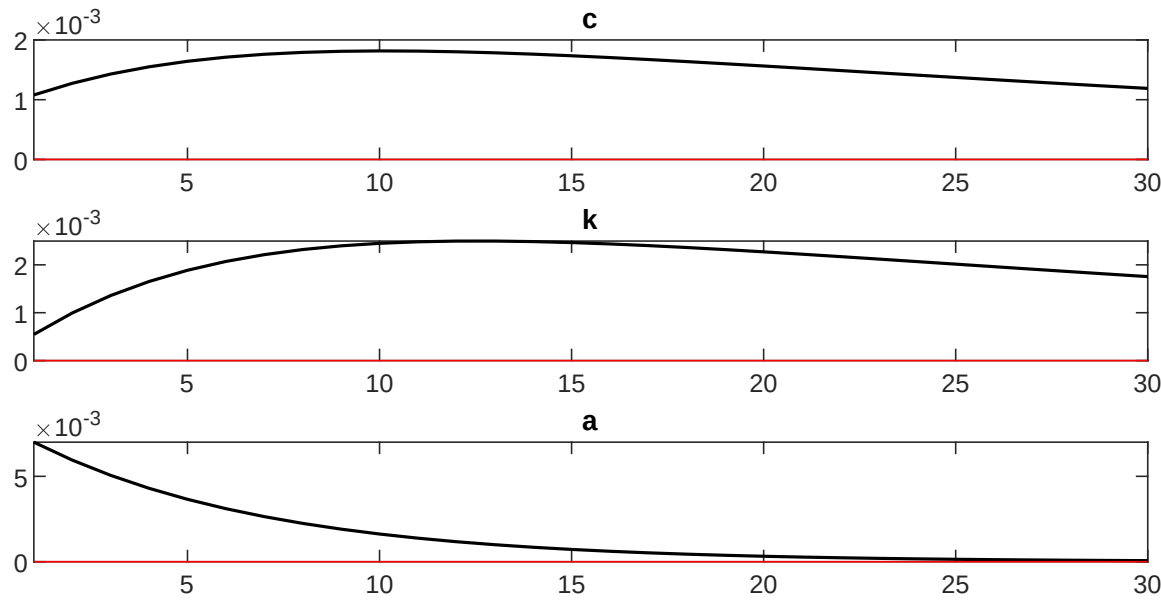
Next, Dynare shows coefficients of autocorrelation up to the fifth order, that is, the correlation of an

endogenous value with p-order lags of itself, $corr(\hat{z}_t, \hat{z}_{t-p})$ where $p = 1, 2, 3, 4, 5$.

console response					
COEFFICIENTS OF AUTOCORRELATION					
Order	1	2	3	4	5
c	0.9965	0.9917	0.9858	0.9788	0.9709
k	0.9992	0.9969	0.9933	0.9884	0.9823
a	0.9500	0.9025	0.8574	0.8145	0.7738

Our model is able to reproduce two simple business cycle facts: (i) the persistence of capital is higher than consumption, (ii) consumption is stable over the business cycles.

Finally, Dynare also plots the system impulse response functions (IRF), *i.e.* the response of endogenous variables after the positive realization of each exogenous shock. Since our model only includes one shock, Equation 3.1 draws the response of consumption $\hat{c}_t$, physical capital $\hat{k}_t$ and productivity $\hat{a}_t$ after the realization of 0.7% shock η_t .



Stored in	Description
<code>oo_</code>	Structure of the output of <code>stoch_simul</code> function
<code>oo_.exo_simul</code>	Simulated series of the exogenous variables (if simulating).
<code>oo_.endo_simul</code>	Simulated series of the endogenous variables (if simulating).
<code>oo_.dr</code>	Stored information about variables and policy function.
<code>oo_.dr.order_var</code>	The mapping from the declaration order to the decision rule order.
<code>oo_.dr.inv_order_var</code>	The mapping from the decision rule order to the declaration order.
<code>oo_.dr.ghx</code>	The A matrix of the transition function.
<code>oo_.dr.ghu</code>	The B matrix of the transition function.
<code>oo_.steady_state</code>	Values of the endogenous variables' steady-state.
<code>oo_.gamma_y</code>	The variance decomposition outputted to the command window.
<code>oo_.mean</code>	The mean of endogenous variables.
<code>oo_.var</code>	The variance-covariance matrix.
<code>oo_.autocorr</code>	The autocorrelation matrices for endogenous variables; up to 5 lags.
<code>oo_.irfs</code>	Impulse responses of endogenous variables to exogenous shocks.
<code>oo_.irfs.x_y</code>	Impulse response of x to an innovation in y.
<code>oo_.forecast</code>	Forecasts of endogenous variables.
<code>M_</code>	Structure with information about the model.
<code>M_.endo_names</code>	The names of endogenous variables in declaration order.
<code>M_.exo_names</code>	The names of exogenous variables in declaration order.
<code>M_.exo_nbr</code>	The number of exogenous variables.
<code>M_.endo_nbr</code>	The number of endogenous variables.
<code>options_</code>	Structure with all the options which Dynare recognises.

Table 1: Matlab's workspace after a stochastic simulation

Exercise 2: extending the simple model

Consider the file [basicRBC.mod](#), the exercise aim at extending the model.

1. The GDP, Y_t and investment I_t are not (yet) included in the model. Include them and draw their responses to a productivity shock. To do so, log-linearize their equations and find their steady state.
2. Check whether the investment- to-gdp ratio $\bar{I}/\bar{Y}$ and $\bar{C}/\bar{Y}$ are consistent with the data (on average, investment and consumption represent respectively 20% and 60% of the GDP for OECD countries). Why is there such a huge gap? How could you improve the RBC model?
3. Change the calibration to get a non-stationnary system response. What is Dynare telling you in the command window? Does increasing return to scale makes the model non-stationnary?

Exercise 3: extension: habits in consumption

Theoretical macro models typically failed at reproducing the excess return premium on financial markets before the 90s. In asset pricing theory, a new consensus emerged to solve these puzzle through the inclusion of new ingredients in macro-models which would help in solving the puzzle. One of them is the consumption habits, pioneered by [Constantinides \[1990\]](#) and [Abel \[1990\]](#). Typically in real life situations, households have strong consumption habits which materializes in time series by having a very stable consumption over the business cycles, compared to other aggregates such as investment and output. Basically, if the household decides to consume more, it will raise the amount of consumption tomorrow. However, there is few evidence about such behaviour at a micro-level, suggesting that consumption habits is probably a shortcut for a more complex explanation of consumption inertia.

1. Let us assume first *external* habits. These external habits simply change how households are enjoying consumption in their utility function:

$$\mathcal{U}(C_{jt} - hC_{t-1}) \quad (25)$$

where $h \in [0, 1)$ is the degree of consumption habits. Here, it is critical to notice that these habits are depending on the aggregate amount of consumption C_{t-1} (i.e. there is no subscript j). So basically, the amount of aggregate consumption is not a control variable when households maximizes utility. This is referred to as *catching up with the joneses* as people will copy the behaviour of the majority of consumers.^a

- (a) The marginal utility of consumption reads as : $\lambda_t = \mathcal{U}'(C_{jt} - hC_{t-1})$.
 - (b) Find the steady state.
 - (c) Linearize the expression. It might be useful to split the problem with $\lambda_t = \mathcal{U}'(x_t)$ and $x_t = C_t - hC_{t-1}$ because the linearization trick cannot be applied for highly non-linear problems.
 - (d) How persistent is now consumption?
2. Let us now assume internal habits, denoted $\mathcal{U}(C_{jt} - hC_{jt-1})$.
 - (a) The marginal utility of consumption reads as : $\lambda_t = \mathcal{U}'(C_{jt} - hC_{jt-1}) - h\beta\mathcal{U}'(C_{jt+1} - hC_t)$.
 - (b) Find the steady state.
 - (c) Linearize the expression. It might be useful to split again the problem with $\lambda_t = \mathcal{U}'(x_t) - \beta h\mathcal{U}'(x_{t+1})$ and $x_t = C_t - hC_{t-1}$.
 - (d) How persistent is now consumption?

^aSee [Abel \[1990\]](#) the story about the Joneses.

Exercise 4: Demand shocks

Assume now that our economy has a government. The government finances public spending by charging a lump-sum tax T_{jt} on households. The total amount of public spending reads as, $P_t G_t$. The balance sheet of government writes:

$$P_t G_t = \int_0^1 T_{jt} dj \quad (=T_t). \quad (26)$$

Since the economy is populated by an uniform household, we can write T_t directly. The budget constraint of the representative household is now:

$$C_{jt} + I_{jt} = A_t K_{jt-1}^\alpha - \frac{T_{jt}}{P_t}, \quad (27)$$

while the resource constraint/the GDP through the expenditure approach is now given by:

$$Y_t = C_t + I_t + G_t. \quad (28)$$

The decision of increasing/decreasing spendings is decided exogenously by the government. In absence of an explicit modelling of political cycles, the spending is assumed to vary around a fixed value of output:

$$G_t = \bar{G} \varepsilon_t^G = g^y \bar{Y} \varepsilon_t^G, \quad (29)$$

where spending follows an $AR(1)$ process:

$$\log(\varepsilon_t^G) = \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G, \quad (30)$$

with $0 \leq \rho_G < 1$ is the autocorrelation coefficient of the demand shock and η_t^G is an innovation distributed normally with zero mean and finite variance σ_G^2 such that $\eta_t^G \sim \mathcal{N}(0, \sigma_G^2)$. Parameter g^y is the spending-to-GDP ratio. We borrow the values of [Smets and Wouters \[2007\]](#) and assume that $\rho_G = 0.97$, $\sigma_G = 0.053$ and $g^y = 0.20$. Implicitly in steady state, the shock is normalized to one $\bar{\varepsilon}^G = 1$.

1. Compute the new steady state of the model. *Tips:* (i) define a new parameter, $g^y = 0.2$, which affects the new steady state; (ii) since the budget is balanced, you can substitute real taxes T_t/P_t by spending G_t .
2. Let's now calibrate the demand shock process, show the IRF after the realization of the demand shock.
3. According to Keynesian economics, is this shock consistent with this theory? Why do households are "Ricardian"? Is this behaviour observed in the data?
4. Analyse the variance decomposition, what is the main driver of output?
5. Is this RBC model now able to replicate another business cycle fact which says output is more volatile than consumption?

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